INDIVIDUAL TASK-04

"Create an ANN Flowchart: Manually sketch a feedforward neural network, label layers, activations, explain how backpropagation adjusts weights**."**

Introduction

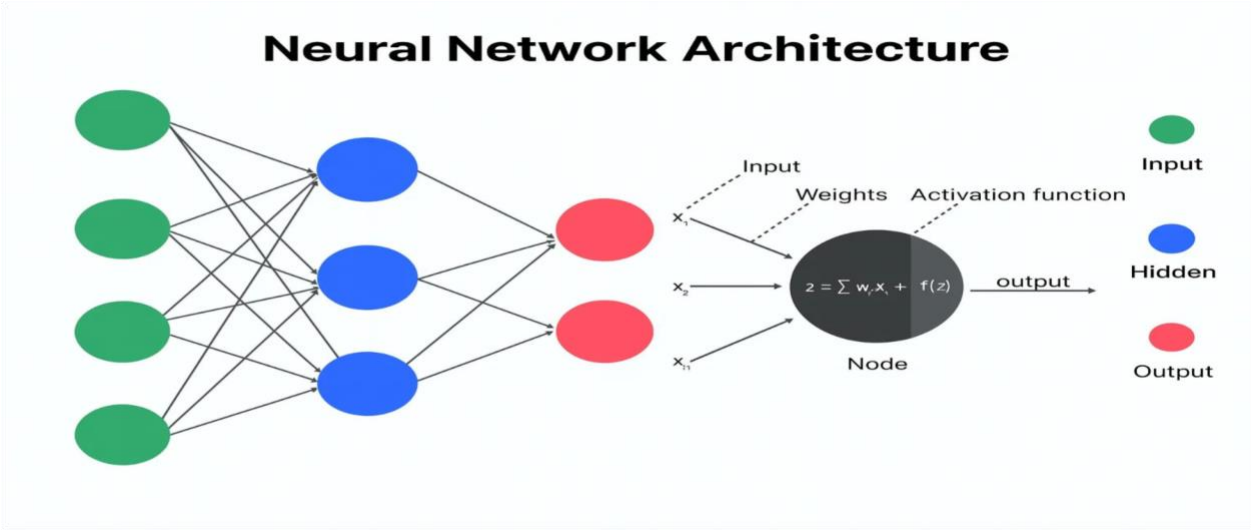
Artificial Neural Network (ANN) is a machine learning model inspired by the structure and functioning of the human brain. Just like the human brain has neurons connected to each other, ANN consists of artificial neurons that are interconnected and work together to process information.

ANN is mainly used to identify patterns, learn from data, and make intelligent decisions. It is widely applied in fields such as image recognition, speech processing, medical diagnosis, and prediction systems.

Architecture of Feedforward Neural Network

A Feedforward Neural Network is the basic type of ANN where the data flows in only one direction, from input layer to output layer, without any loops or cycles.

The diagram illustrates a standard **Feedforward Neural Network**. Every circle represents a "neuron" or "node."



➢ Input Layer (Detailed)

The input layer is the first layer of the neural network. It is responsible for receiving the input data from the external environment. Each neuron in this layer represents one feature of the dataset.

For example, if we are predicting house prices, inputs can be size, location, number of rooms, etc. The input layer does not perform any computation; it simply passes the data to the next layer.

First layer of the network
Receives input data (features like X1, X2, X3)
Passes data to the next layer

➢ Hidden Layer(s)

Hidden layers are placed between input and output layers. These layers perform the actual computation and learning process.The presence of multiple hidden layers helps the network to learn complex patterns and relationships in the data. Deep neural networks may contain many hidden layers. The hidden layer is where the actual "learning" happens. In your diagram, these are the **Blue Nodes**. .

Each neuron in the hidden layer:

- Receives input from previous layer
- Multiplies input with weights
- Adds bias
- Applies activation function

➢ Output Layer (Detailed)

The output layer is the final layer of the network. It provides the final result or prediction. The final layer that provides the prediction (e.g., "This is a cat" or "Price = $200")ˇ The output layer (the **Red Nodes** in your image) is the final stage of the feedforward process. It translates all the internal mathematical calculations into a format that humans can understand. The type of output depends on the problem:

- Classification → gives class label (Yes/No, Spam/Not Spam)
- Regression → gives numeric value (price, temperature

➢ Weights and Bias

In an Artificial Neural Network (ANN), **weights and bias** are the most important parameters that control how the network learns and makes predictions. They determine the strength and behavior of connections between neurons.

Weights and biases are very important parameters in ANN.

- **Weights:** Determine the importance of input features
- **Bias:** Helps shift the activation function output

Initially, weights are assigned randomly. During training, they are updated continuously to improve accuracy.

Z=(W1X1+W2X2+W3X3)+b

➢ Activation Function

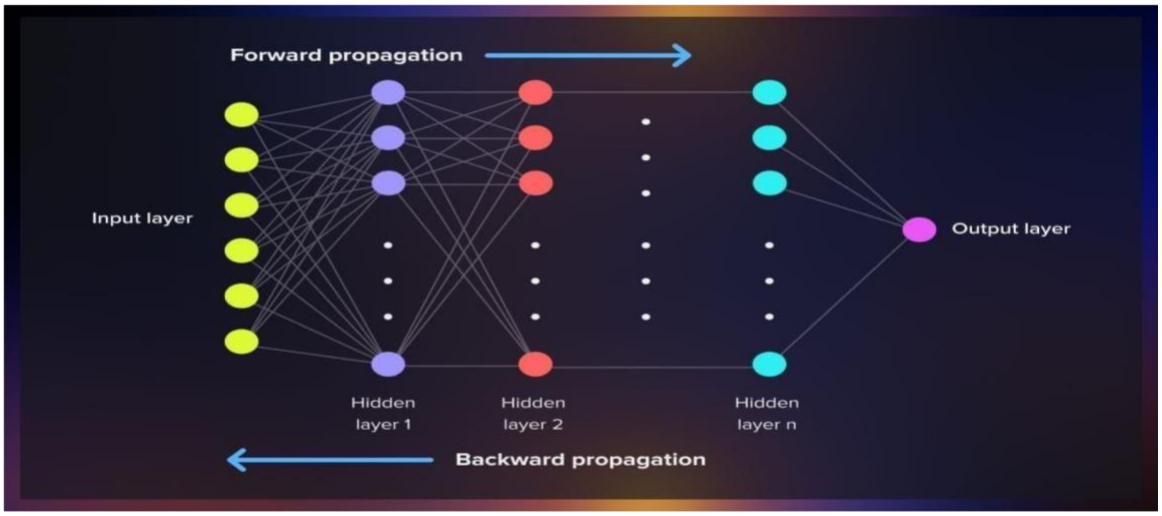
Activation functions are mathematical functions applied to the output of each neuron. They introduce non-linearity, allowing the network to learn complex patterns.

Common Activation Functions:

- **ReLU (Rectified Linear Unit):** Outputs zero for negative values and input itself for positive values. It is widely used because it is simple and efficient.
- **Sigmoid Function:** Converts output into values between 0 and 1. Mainly used in binary classification.
- **Tanh Function:** Outputs values between -1 and 1.

FORWARD PROPAGATION

Forward propagation in neural networks refers to the process of passing input data through the network's layers to compute and produce an output. Each layer processes the data and passes it to the next layer until the final output is obtained. During this process, the network learns to recognize patterns and relationships in the data, adjusting its weights through backpropagation to minimize the difference between predicted and actual outputs.



Step-by-StepWorking

➢ Input Layer Processing

The input layer receives the initial data from the external environment. Each neuron in this layer represents a feature of the dataset. The input layer does not perform any computation; it only forwards the data to the hidden layer.Input features (X1, X2, X3, …) are fed into the network These values are passed to the hidden layer **.**

➢ Weighted Sum Calculation

Each neuron in the hidden layer receives input values and multiplies them with corresponding weights. These weighted inputs are then summed together to produce a single value. This step determines how much influence each input has on the neuron.

Formula (you can write in exam):

$Z=(W1X1+W2X2+W3X3)+b$

➢ Activation Function Application

The result of the weighted sum and bias is passed through an activation function. This function introduces non-linearity into the model, enabling it to learn complex patterns. Without activation functions, the neural network would behave like a simple linear model.

Example:

- ReLU → removes negative values
- Sigmoid → converts output between 0 and 1

➢ Passing Through Hidden Layers

The hidden layers perform multiple transformations on the input data. Each hidden layer extracts important features and patterns from the data. The deeper the network, the more complex features it can learn.

- Output from one layer becomes input to next layer
- Same process repeats (weights + bias + activation)

➢ Layer-by-Layer Computation

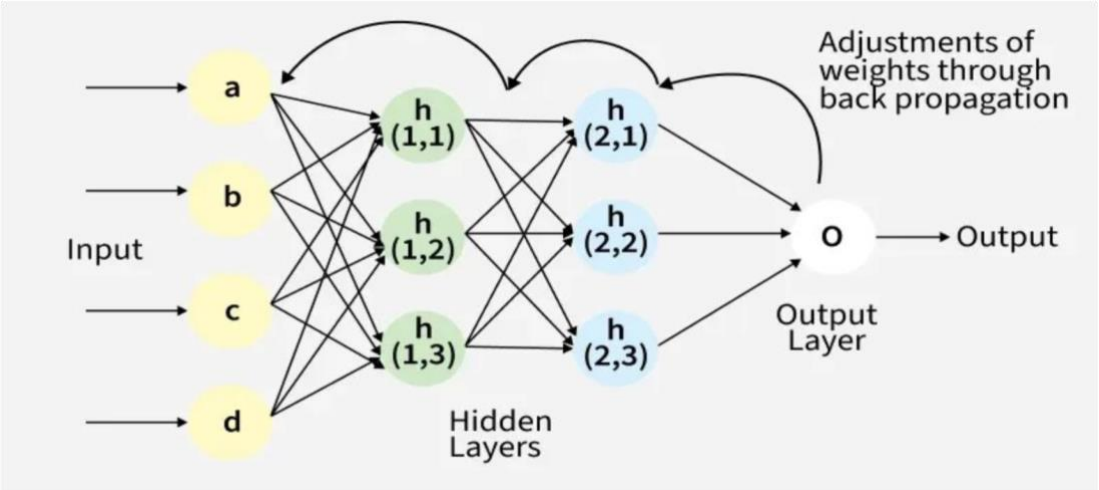
Forward propagation happens layer by layer. The output of one layer becomes the input to the next layer. This continues until the final layer is reached, ensuring smooth data processing throughout the network.

- Final output is generated
- This is the predicted value (ŷ)

Backpropagation in Neural Network

Backpropagation is essentially a massive application of the **Chain Rule** from calculus. Because a neural network is a nested function—where the output of one layer is the input of the next—we cannot change a weight in the first layer without knowing how it affects the final output.

- **The Problem:** We know the "Total Error" at the end, but we don't know which specific weight (w) in the middle of the network caused it.
- **The Solution:** The Chain Rule allows us to calculate the **Gradient** ($\frac{\partial Error}{\partial w}$). It breaks down the complex error into small pieces, telling us: "If I change this weight by 0.01, the total error will drop by 0.05."



Multi-Layer Perceptron (MLP) Architecture

The image displays a **Deep Neural Network** because it has more than one hidden layer ($h1$ and $h2$).

- **Input Layer (Nodes a, b, c, d):** These represent the independent variables. If we were predicting house prices, these might be "Square Footage," "Location," "Number of Rooms," and "Age."
- **Hidden Layer 1 (Green Nodes):** These nodes receive signals from all four inputs. This is where the network creates **low-level abstractions** (simple patterns).
- **Hidden Layer 2 (Blue Nodes):** These nodes receive signals from the green layer, creating **high-level abstractions** (complex patterns).
- **Fully Connected (Dense) Structure:** Notice how every node in one layer is connected to every node in the next. This ensures the network can find relationships between any two pieces of information.

Backpropagation – The "Feedback Loop"

The **curved arrows** at the top are the most important part of your image. They represent **Backpropagation**.

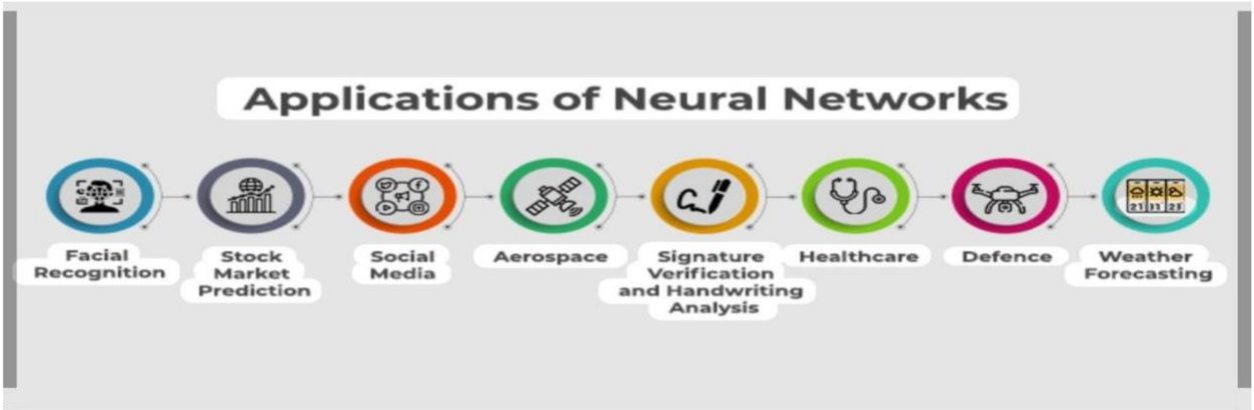
- **Error Detection:** Once the signal reaches the **Output Layer (O)**, the network realizes there is a gap between its prediction and the truth. This gap is the **Loss**.
- **Credit Assignment:** The curved arrows show the "Error Signal" traveling backward. The network is asking: *"How much of this error was caused by the blue nodes? How much by the green nodes?"*
- **The Gradient:** The network calculates the **Gradient** (slope) of the error for every single weight in the system.

The label ***"Adjustments of weights through backpropagation"*** refers to the optimization step.

- Gradient Descent: The network uses the information from the curved arrows to update the weights.
- Convergence: By repeating this "Forward $\rightarrow$ Backward $\rightarrow$ Adjust" cycle hundreds of times (epochs), the weights eventually reach their optimal values. At this point, the network has "learned" the task.

Real-World Applications of Artificial Neural Networks

Artificial Neural Networks (ANNs) are not just theoretical; they power the modern world. Because of their ability to learn through **Backpropagation** (as shown in your previous diagrams), they can solve problems that are too complex for traditional programming.



The beauty of Neural Networks lies in their versatility. Because they can learn from any numerical data through **Backpropagation**, they have moved from the lab into these 9 critical real-world fields:

Facial Recognition

- **The Tech:** Uses Convolutional Neural Networks (CNNs).
- **Process:** The network identifies unique spatial features (distance between eyes, nose shape) to verify identity.
- **Use Case:** Unlocking smartphones, airport security, and law enforcement.
- **Stock Market Prediction**
- **The Tech:** Uses Recurrent Neural Networks (RNNs) or LSTMs.
- **Process:** These networks have a "memory" of time-series data, allowing them to spot trends in historical price movements to predict future fluctuations. ⍰
- **Use Case:** Automated trading bots an

Social Media

- **The Tech:** Deep Learning Recommendation Engines.
- **Process:** Networks analyze your likes, watch time, and friends' activity to predict what content will keep you engaged.
- **Use Case:** Instagram Explore page, TikTok's "For You" feed, and Facebook ads.

Conclusion

Artificial Neural Networks are one of the most powerful tools in machine learning and artificial intelligence. They mimic the working of the human brain and are capable of learning from data automatically. With the help of forward propagation and backpropagation, ANN continuously improves its performance by adjusting weights and reducing errors. Due to their ability to solve complex problems, ANNs are widely used in modern AI applications and continue to play a significant role in technological advancements.

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Key Takeaways:

- ANN mimics the human brain with neurons, layers, weights, and biases.
- Forward propagation generates output using current weights and bias.
- Backpropagation adjusts weights and bias to reduce error.
- Activation functions introduce non-linearity, enabling complex learning.
- This combined process makes ANN a versatile and powerful tool in AI and ML.

INDIVIDUAL TASK-05

"Chatbot Sales Conversation: Interact with a customer-service bot. Analyze how well it interprets intent and closes a "deal."”

Introduction

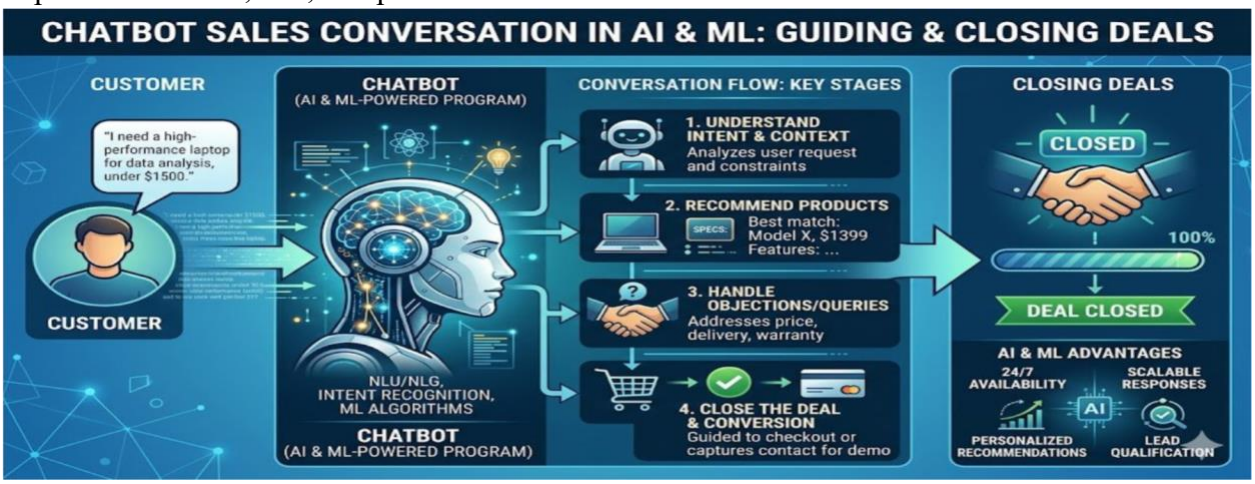
A **chatbot** is an AI-powered program designed to simulate human conversation. In **sales**, chatbots function as **virtual sales assistants**, guiding customers, answering queries, recommending products, and even helping close deals. By using **Artificial Intelligence (AI)**, **Machine Learning (ML)**, and **Natural Language Processing (NLP)**, chatbots can understand customer intent, provide personalized suggestions, and improve their performance over time.

By leveraging **Machine Learning (ML)** and **Natural Language Processing (NLP)**, chatbots can **understand customer intent**, recommend products, handle objections, and improve their responses over time. Chatbot sales conversations have become an important tool in modern business to **increase efficiency, reduce costs, and enhance customer experience**.

A **chatbot** is a computer program designed to simulate human conversation using **Artificial Intelligence (AI)**. In the context of **sales**, chatbots act as **virtual assistants** that interact with customers, provide information, answer queries, and even help in **closing deals**.

Chatbot Sales Conversation in AI & ML

A **chatbot** is an AI-powered program that interacts with users using natural language. In sales, chatbots can guide customers, answer queries, recommend products, and even close deals. These bots are designed to simulate human conversation in a seamless way, making the customer experience smooth, fast, and personalized.



- ### AI and ML in Sales Conversations:

- **Intent Interpretation (The "Brain"):** This is the ability to identify the *purpose* behind a user's message. In sales, this means distinguishing between "Just looking" (low intent) and "How much does it cost?" (high intent).
- **Closing the Deal (The "Action"):** In AI terms, this is often called **Target-Driven Dialogue**. The model doesn't just respond; it uses reinforcement learning to select the response that has the highest statistical probability of leading to a "conversion event" (like booking a demo or adding to cart).

- ## Definition and Purpose

A chatbot is an AI-powered program designed to simulate human conversation using text or voice. In sales, it acts as a virtual assistant that interacts with customers, answers queries, recommends products, and even helps close deals.

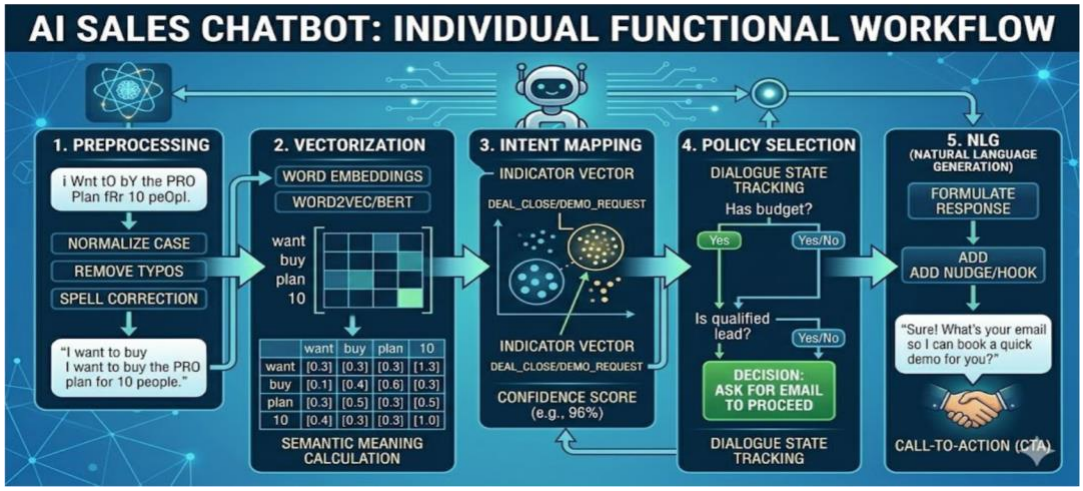
- To provide instant support to customers.
- To understand customer intent and requirements.
- To **recommend products or services** based on customer queries.

Steps in a Chatbot Sales Conversation

- **Greeting the Customer** :Bot welcomes the user politely.
  - Example: "Hello! I'm your virtual assistant. How can I help you today?"
- **Understanding Customer Intent** :Bot uses **NLP** to analyze the query.
  - Determines if the customer wants product info, price, recommendations, or checkout help.
- **Providing Information:** Bot presents relevant details about products/services.
  - Uses **ML** to personalize suggestions based on past interactions or preferences.
- **Handling Queries** : Resolves objections like pricing, availability, or product comparison.
- **Guiding the Sales Funnel** :Suggests products, bundles, or promotions. Encourages the customer to take action: "Add to Cart" or "Buy Now".
- **Closing the Deal** :Confirms order details and payment options. Provides checkout links and post-sale support info.

The Functional Workflow

In the context of an AI-powered sales chatbot, the workflow serves as the step-by-step operating system that translates messy human language into structured, goal-oriented business actions.Its primary function is to create a structured path that moves a user from a vague query to a specific, completed transaction or business goal.



	want	buy	plan	10
want	[0.3]	[0.3]	[0.3]	[1.3]
buy	[0.1]	[0.4]	[0.6]	[0.3]
plan	[0.3]	[0.5]	[0.3]	[0.5]
10	[0.4]	[0.3]	[0.3]	[1.0]

- **Input Normalization (Data Cleaning)**

The Function: This initial stage takes user input (e.g., "i Wnt tO bY the PRO Plan fr 10 peOpl." from image_2.png) and cleans it. By removing typos and normalizing capitalization, the workflow ensures the following AI models are analyzing clean, consistent data, which significantly increases accuracy.

- **Meaning Extraction (Semantic Understanding)**

The Function: Since machines cannot interpret words directly, the workflow converts text into numerical vectors (as seen in the "VECTORIZATION" step in image_2.png). The function here is to mathematically represent the semantic meaning and relationship of words, allowing the system to "calculate" the intent behind the user's message.

- **Intent Classification and Identification**

The Function: The central function of this stage is pattern matching. The workflow compares the current user's vectorized query against pre-defined or trained clusters of known user intents (e.g., matching a query to the Request_Demo cluster in image_2.png with a high confidence score). This tells the bot exactly what the user is trying to accomplish.

Advantages of Chatbot Sales Conversations

1. **Handles Multiple Customers** – A single chatbot can engage with hundreds or thousands of users simultaneously.
2. **Cost-Effective** – Reduces the need for a large human sales team while providing consistent service.

3. **Personalized Customer Experience** – Chatbots use data from past interactions to suggest products tailored to individual preferences.

4. **Data Collection and Analytics** – They collect conversation data, helping businesses understand customer behavior and improve marketing strategies.

5. **Increased Sales Conversion** – By guiding customers through the buying process and providing timely suggestions, chatbots can boost sales.

6. **Reduces Workload for Human Agents** – Human agents can focus on complex queries while chatbots handle repetitive tasks.

Disadvantages / Challenges of Chatbot Sales Conversations

1. **Limited Understanding** – Chatbots may misinterpret complex or ambiguous queries.

2. **Lack of Human Empathy** – They cannot fully understand emotions or respond like a human in sensitive situations.

3. **Requires Constant Training** – To maintain accuracy, chatbots need continuous updates and machine learning improvements.

4. **Limited in Handling High-Value Sales** – For big-ticket items or negotiations, human interaction is often preferred.

5. **Dependence on Pre-Defined Scripts** – Some chatbots may fail if the conversation deviates from their training or scripts.

6. **Technical Issues** – Bots can face glitches or downtime, which may frustrate customers.

Key Components

Phase A: Intent & Entity Recognition

- **Intent Classification:** Categorizing the user's goal (e.g., Check_Price, Request_Demo, Handle_Objection).
- **Entity Extraction:** Identifying the "variables" of the deal (e.g., Product Name: "Pro Plan", Quantity: "50 seats", Budget: "$5000").
- **Confidence Scoring:** The bot assigns a percentage (e.g., 0.92) to how sure it is of your intent. If it's too low, it asks a "clarifying question" instead of guessing.

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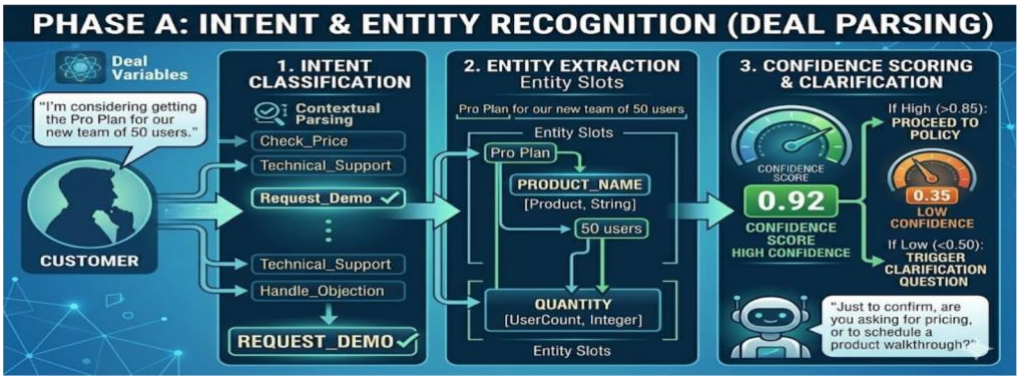
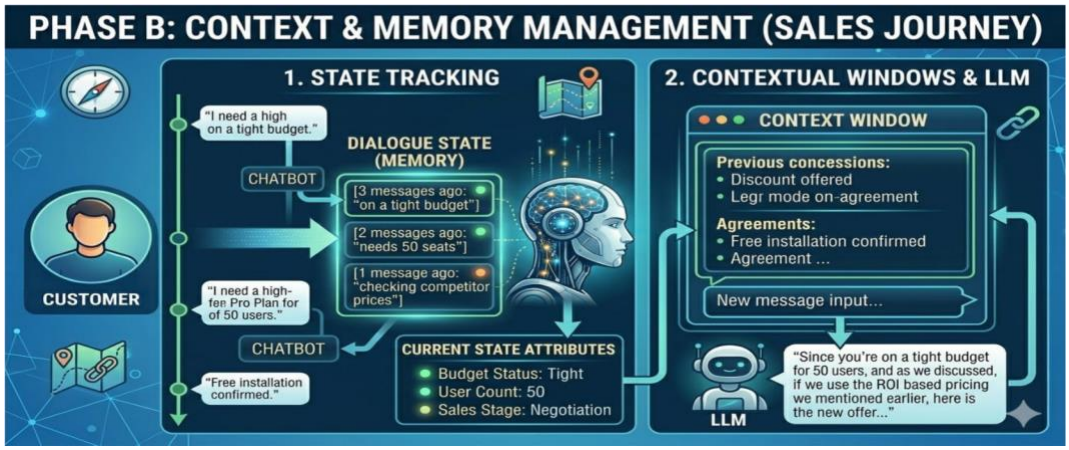


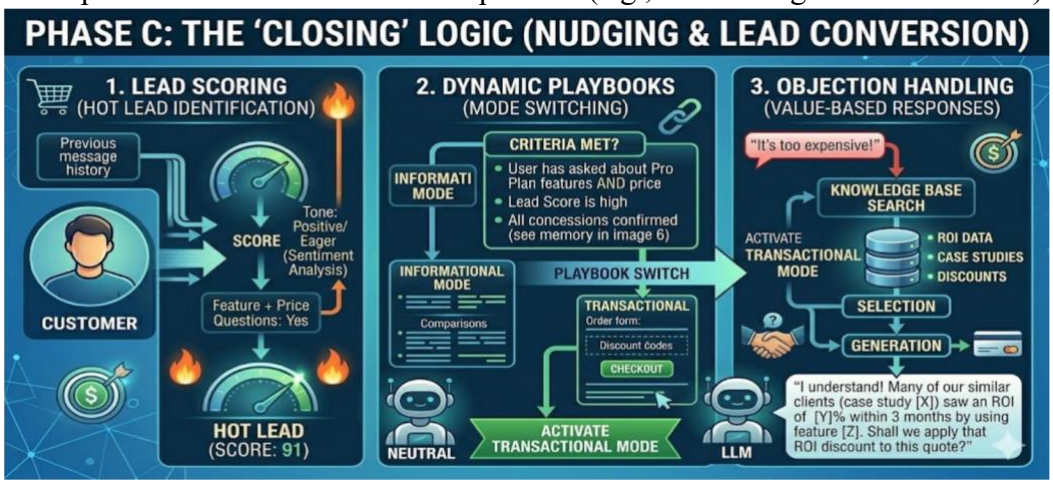
Phase B: Context & Memory Management

- **State Tracking:** The AI keeps a "Dialogue State" to remember that three messages ago, you said you were "on a tight budget."
- **Contextual Windows:** Modern LLMs use a context window to ensure the current response aligns with previous concessions or agreements made during the chat.



Phase C: The "Closing" Logic (Nudging)

- **Lead Scoring:** The ML model analyzes your tone (Sentiment Analysis) and the complexity of your questions to score how "hot" the lead is.
- **Dynamic Playbooks:** The bot switches from "Informational Mode" to "Transactional Mode" once specific criteria are met (e.g., the user has asked about both features and price).
- **Objection Handling:** Using predefined knowledge bases to counter "It's too expensive" with "Value-based responses" (e.g., mentioning ROI or discounts).



KeyAI/MLComponentsinChatbots:

- **Intent Recognition**: Understands what the user wants.
- **Entity Recognition**: Extracts specifics like product names, colors, sizes, or price range.
- **Dialogue Management**: Determines the next step in conversation.
- **Recommendation Engine**: Suggests products based on user behavior or preferences.
- **Sentiment Analysis**: Detects emotions such as satisfaction, confusion, or frustration.

Evaluation Metrics:

- **Goal Completion Rate (GCR):** Did the user reach the end of the sales funnel?
- **Turns to Goal:** How many messages did it take? (Fewer is usually better in sales).
- **Fall-back Rate:** How often did the bot say "I don't understand"? (High rates kill deals).

Conclusion

Chatbot sales conversations are a **powerful application of AI and ML** in modern business. By using **Artificial Intelligence, Machine Learning, and Natural Language Processing**, chatbots can simulate human-like interactions, understand customer intent, and provide **personalized product recommendations**. They help businesses improve efficiency by handling multiple customers simultaneously, offering **24/7 support**, and reducing workload for human sales teams. While chatbots have **many advantages**—such as instant responses, cost-effectiveness, and increased sales conversion—they also face **limitations**, including difficulty in understanding complex queries and lack of human empathy. Despite these challenges, with continuous training and integration with AI/ML tools, chatbots can **learn from past interactions** and become more accurate and persuasive over time.

In summary, chatbots enhance customer engagement, streamline the sales process, and are an essential tool for businesses looking to **combine technology with effective sales strategies**.

- Chatbots are **AI-powered virtual assistants** for sales.
- They use **AI, ML, and NLP** to understand customer intent and respond effectively.
- Help in **personalized product recommendations** and **guiding customers** through the buying process.
- **Reduce workload** for human agents and are **cost-effective**.
- **Collect data** to improve future interactions and marketing strategies.
- **Limitations** include inability to handle complex queries fully and lack of human empathy.
- With proper training and ML updates, chatbots **learn and improve over time**, enhancing sales efficiency.